JUNIOR SCHOOL LEARNING HUB

GRADE 8 INTEGRATED SCIENCE

DETAILED TEACHING NOTES • FULL YEAR • 2027

School: _____ **Teacher:** _____

TSC No.: _____ **Learning Area:** Integrated Science

Curriculum basis: Grade 8 Integrated Science curriculum design (KICD, revised 2024). These are school-neutral teacher/learner support notes prepared to explain the listed Grade 8 substrands. They are not an official KICD publication.

How to use these notes

• Use the notes alongside the current KICD curriculum design, approved learner's book and the school's scheme of work.

• Select explanations, demonstrations and practice tasks that match the lesson outcome being taught.

• Practical activities should be supervised. Use safe, locally available materials and follow school laboratory rules.

• Assessment questions are formative practice; teachers should adapt them to learner readiness and the exact curriculum emphasis.

1.1 Elements and Compounds

This substrand introduces substances and the way elements combine to form compounds. Learners should use observable evidence, models and simple representations to distinguish elements, compounds and mixtures.

Matter and pure substances

Matter is anything that has mass and occupies space. A pure substance has a fixed composition and characteristic properties. Elements and compounds are pure substances; mixtures contain two or more substances physically combined in variable proportions.

Elements

• An element is a pure substance made of only one type of atom.

• Elements cannot be broken down into simpler substances by ordinary chemical methods.

• Each element is represented by a chemical symbol. The first letter is capitalised; a second letter, where present, is lower case (for example H, O, Na and Cl).

• Elements may be metals, non-metals or metalloids. Examples include iron, copper, oxygen, carbon and silicon.

• Atoms of an element have the same number of protons. Atoms may differ in the number of neutrons.

Compounds

• A compound is a pure substance formed when atoms of two or more different elements are chemically combined in a fixed ratio.

• The properties of a compound are generally different from the properties of the elements that formed it.

• Water (H■O) contains hydrogen and oxygen chemically combined; sodium chloride (NaCl) contains sodium and chlorine.

• A compound has a definite chemical formula. The formula shows the kinds of atoms and their relative numbers.

• Compounds can be separated into simpler substances only by chemical processes, not by ordinary physical separation.

Elements compared with compounds and mixtures

Feature	Element	Compound	Mixture
Composition	One type of atom	Two or more elements chemically combined	Two or more substances physically combined
Proportion	One element	Fixed ratio	Variable ratio
Separation	Not by ordinary chemical methods	Chemical methods	Physical methods
Properties	Characteristic of element	Different from constituent elements	Components retain many of their properties

Models and representations

• Use coloured counters, beads or paper circles to represent atoms. A single colour can represent one element; joined different colours can represent a compound molecule.

• A model is a simplified representation. It helps explain composition but is not a picture of the actual size, colour or arrangement of atoms.

• Learners can draw particle diagrams and write simple formulae where required by the curriculum and teacher guidance.

Suggested learning activities

• Sort labelled samples or pictures into elements, compounds and mixtures, explaining the evidence used.

• Build simple particle models using counters and compare a model of an element with a model of a compound.

• Interpret familiar formulae such as H■O, CO■ and NaCl, identifying the elements represented.

• Discuss where elements and compounds are encountered at home, in school, in agriculture and in industry.

Check your understanding

• Define an element and a compound.

• Give two examples of elements and two examples of compounds.

• Explain why water is a compound and air is a mixture.

• State two differences between a compound and a mixture.

• What information is communicated by the formula CO■?

1.2 Physical and Chemical Changes

Learners investigate changes in materials and use evidence to distinguish physical changes from chemical changes. Emphasise observation, fair testing, recording and safe handling.

Physical changes

• A physical change alters the form, state, size or appearance of a substance without producing a new substance.

• Examples include melting ice, freezing water, dissolving sugar in water, cutting paper and breaking a piece of chalk.

• Many physical changes can be reversed, but reversibility alone is not a reliable test: some physical changes are difficult to reverse.

• During a change of state, the substance remains the same. Ice, liquid water and water vapour are all water.

Chemical changes

• A chemical change produces one or more new substances with properties different from those of the starting materials.

• Examples include rusting of iron, burning wood, cooking an egg and reaction of vinegar with baking soda.

• Chemical changes often involve energy transfer. Heat, light or sound may be released or absorbed.

• Possible evidence includes a permanent colour change, gas production, formation of a precipitate, temperature change or light emission. No single observation proves a chemical change in every case; interpret evidence together.

Comparing the changes

Question	Physical change	Chemical change
New substance formed?	No	Yes
Composition	Remains the same	Changes as atoms rearrange
Examples	Melting, freezing, cutting	Rusting, burning, cooking
Reversal	May be easy or difficult	Often difficult by simple physical means

Energy in changes

• Changes may be accompanied by energy transfer. In an exothermic process, energy is transferred to the surroundings, often warming them. In an endothermic process, energy is taken in from the surroundings, often

cooling them.

• Burning fuels releases energy to the surroundings. Melting requires energy to be transferred to the substance.

• Temperature observations should be made with suitable apparatus and without touching hot materials.

Investigation: physical and chemical changes

• Observe safe examples such as ice melting, paper being cut, sugar dissolving and an approved teacher demonstration of a chemical reaction.

• Record the starting materials, observations during the change and final material.

• Decide whether a new substance is likely to have formed and justify the decision using evidence.

• Keep one variable changed at a time when comparing conditions in an investigation.

Safety and environmental links

• Never taste laboratory materials. Do not directly smell chemicals; use teacher-approved wafting only when instructed.

• Do not burn unknown materials. Any flame demonstration must be conducted by the teacher in a suitable safe area.

• Discuss how corrosion, combustion, food preparation and waste treatment affect daily life and the environment.

Check your understanding

• Explain the difference between a physical and a chemical change.

• Classify melting candle wax, rusting iron and dissolving salt, giving a reason for each.

• Give three observations that may indicate a chemical change.

• Why is reversibility not the only test for identifying a physical change?

• Describe a safe investigation for comparing two changes.

1.3 Classes of Fire

This substrand develops understanding of fire, fire prevention and safe responses. Fire is dangerous; the main classroom emphasis is prevention, recognition and safe action, not learner operation of extinguishers or making fires.

The fire triangle

• Fire requires three conditions: heat, fuel and oxygen. These are commonly represented by the fire triangle.

• Removing any one of the three conditions can stop combustion. Cooling removes heat; separating fuel removes the fuel source; smothering can reduce oxygen.

• Fire prevention includes safe storage of fuels, careful use of electrical equipment, keeping ignition sources away from flammable materials and maintaining clear exits.

Fuel and ignition

• A fuel is a material that can burn. Examples include paper, dry wood, cooking gas and some liquids.

• An ignition source supplies enough energy to start combustion. Examples include a flame, spark or hot surface.

• Different fuels and situations require different control measures. Never investigate ignition using learner-made flames.

Common fire classes

• Fire classifications are based on the type of material burning. The letter labels and extinguisher colour coding can vary by national standard; teachers should use the current locally approved fire-safety chart and equipment labels.

• Common categories in many systems include ordinary solid combustibles such as wood, paper and cloth; flammable liquids; flammable gases; combustible metals; and cooking oils/fats.

• Electrical equipment that is energised presents an electric-shock hazard. Follow the equipment label and local fire-safety guidance; do not apply water to live electrical equipment.

• Learners should identify the fuel involved from safe pictures, case studies or teacher explanations rather than approaching a real fire.

Choosing a safe response

• Raise the alarm, alert responsible adults and emergency services, and evacuate using the planned route.

• Do not hide, return to collect belongings, or attempt to fight a fire unless trained and authorised.

• Water can spread burning liquids and can conduct electricity; therefore it is unsuitable for some fires.

• Only trained adults should select and operate extinguishers, following the extinguisher label and local safety procedures.

Fire prevention at home and school

• Keep matches, lighters and fuels away from children and heat sources.

• Avoid overloading sockets and report damaged electrical cables.

• Keep corridors, stairways and exits clear.

• Store chemicals and fuels in approved, labelled containers away from ignition sources.

• Know the assembly point and participate seriously in fire drills.

Learning activities and assessment

• Interpret a fire-triangle diagram and explain how removing one condition affects combustion.

• Classify pictured fire scenarios by the material burning, using a locally approved chart.

• Prepare a school fire-prevention checklist and identify safe evacuation actions.

• Role-play raising an alarm and moving calmly to an assembly point; do not simulate a real fire.

Check your understanding

• Name the three conditions in the fire triangle.

• Explain two ways of preventing fires at school.

• Why should water not be used on burning cooking oil or live electrical equipment?

• State four safe actions to take when a fire is discovered.

• Why should learners not attempt to operate an extinguisher without training?

2.1 The Cell

The cell is the basic structural and functional unit of living organisms. Learners observe prepared slides, images or models and relate cell structures to their functions.

Cell theory and cell size

• All living organisms are made of one or more cells; the cell is the basic unit of structure and function; and new cells arise from existing cells.

• Most cells are too small to be seen clearly with the unaided eye. A microscope makes small structures appear larger.

• Cells differ in shape and size according to their roles.

Main parts of a typical cell

• The cell membrane forms the boundary of the cell and controls movement of substances into and out of it.

• The cytoplasm is the jelly-like region where many cell activities take place.

• The nucleus contains genetic material and helps control cell activities.

• Plant cells also have a cell wall made mainly of cellulose, which supports and strengthens the cell.

• Many plant cells contain chloroplasts with chlorophyll for photosynthesis. A large permanent vacuole contains cell sap and helps maintain firmness.

Plant and animal cells

Feature	Typical plant cell	Typical animal cell
Cell membrane	Present	Present
Cell wall	Present	Absent
Nucleus	Usually present	Usually present
Chloroplasts	Present in many green plant cells	Absent
Large permanent vacuole	Usually present	Usually absent

Using a microscope

• Carry the microscope with two hands and place it on a stable bench.

• Prepare or use a teacher-provided slide. Begin with the low-power objective lens.

• Adjust illumination and focus carefully; use the coarse adjustment only at low power and fine adjustment for clearer focus.

• Draw what is actually observed, using clear lines and labels. Do not add structures that cannot be seen.

• Clean lenses only with lens tissue and follow teacher instructions.

Specialised cells

• Cells may be adapted to perform particular functions. Examples include red blood cells for oxygen transport, nerve cells for carrying impulses, root hair cells for absorption and guard cells for controlling stomata.

• A cell's shape and structures are related to its function. Use labelled diagrams and comparisons to explain these adaptations.

Check your understanding

• State the cell theory in three points.

• Name the main parts of an animal cell and state their functions.

• Give three structures found in a typical plant cell that are absent from an animal cell.

• Explain why a microscope is useful in studying cells.

• Relate the structure of a root hair cell to its function.

2.2 Movement of Materials In and Out of the Cell

This substrand explains how substances move across the cell membrane. Use particle models, simple investigations and familiar examples to develop understanding.

Cell membrane and exchange

• Cells exchange substances with their surroundings. Oxygen and dissolved nutrients enter cells; carbon dioxide and other wastes leave cells.

• The cell membrane is selectively permeable: it allows some substances to pass more easily than others.

• Movement across membranes helps cells maintain suitable internal conditions.

Diffusion

• Diffusion is the net movement of particles from a region of higher concentration to a region of lower concentration, due to random particle motion.

• Diffusion continues until particles are more evenly distributed, although particles keep moving.

• Examples include perfume smell spreading through a room and oxygen moving from the blood into body cells.

• Diffusion is faster when temperature is higher, the concentration difference is greater, or the distance for movement is shorter.

Osmosis

• Osmosis is the net movement of water molecules through a selectively permeable membrane from a region of higher water concentration (more dilute solution) to a region of lower water concentration (more concentrated solution).

• A selectively permeable membrane is essential to the definition of osmosis.

• Plant cells in dilute solutions may become firm as water enters. In concentrated solutions, water may leave and the cell contents may shrink away from the cell wall.

• Osmosis helps explain water uptake by roots and movement of water between cells and their surroundings.

Active transport

• Active transport moves substances across a membrane against a concentration gradient, from lower concentration to higher concentration.

• It requires energy released by cellular respiration and uses specific membrane transport proteins.

• An example is uptake of mineral ions by root hair cells when the concentration in the soil is lower than inside the cells.

Simple investigations

• Place equal-sized potato pieces in solutions of different concentrations under teacher guidance; record changes in mass, length or firmness.

• Use a safe coloured solution or food colouring to model diffusion in water.

• Identify the variable changed, variables kept constant and the observations recorded.

• Do not taste any solution or use household chemicals without teacher approval.

Check your understanding

• Define diffusion and osmosis.

• Explain why osmosis requires a selectively permeable membrane.

• Distinguish diffusion from active transport.

• Predict what may happen to a plant cell placed in a very concentrated solution.

• Give one example of diffusion and one of active transport in living organisms.

2.3 Reproduction in Human Beings

These notes support age-appropriate, respectful and scientifically accurate learning about human reproduction, puberty and reproductive health. Teach sensitively, protect learner privacy and follow school safeguarding procedures.

Reproduction and its importance

• Reproduction is the biological process by which living organisms produce offspring. In humans, reproduction is sexual and involves male and female reproductive cells (gametes).

• The male gamete is the sperm cell; the female gamete is the ovum (egg cell). Fertilisation is the joining of the nuclei of the sperm and ovum to form a zygote.

Male reproductive system

• The testes produce sperm cells and the hormone testosterone.

• The scrotum holds the testes outside the main body cavity and helps maintain a suitable temperature for sperm production.

• Sperm travel through ducts and are released through the urethra in semen. Accessory glands add fluids that nourish and transport sperm.

• Use a properly labelled, age-appropriate diagram; explain structures and functions without embarrassment or stigma.

Female reproductive system

• The ovaries produce ova and hormones including oestrogen and progesterone.

• The oviducts (fallopian tubes) carry the ovum; fertilisation commonly occurs in an oviduct.

• The uterus is where an embryo implants and develops during pregnancy. The cervix is the lower opening of the uterus, and the vagina connects the cervix to the outside of the body.

• The menstrual cycle involves changes in the ovaries and uterus. Menstruation occurs when the uterine lining is shed if pregnancy has not occurred.

Puberty

• Puberty is the period when a child's body develops towards sexual maturity under the influence of hormones.

• Changes vary between individuals and occur at different times. They may include growth spurts, changes in body shape, development of body hair, skin changes and emotional changes.

• Menstruation and sperm production are biological developments associated with reproductive maturity.

• Personal hygiene, correct health information, respectful relationships and seeking guidance from a trusted adult are important.

Fertilisation, pregnancy and birth

• During fertilisation, a sperm nucleus fuses with an ovum nucleus to form a zygote.

• The zygote divides repeatedly and develops into an embryo. Implantation occurs when the early embryo attaches to the lining of the uterus.

• The placenta allows exchange of oxygen, nutrients and waste products between maternal and fetal blood without the two blood supplies normally mixing directly.

• The umbilical cord connects the developing baby to the placenta. Pregnancy ends with birth.

Reproductive health and respect

• Maintain personal hygiene and use accurate scientific information.

• Respect privacy, bodily boundaries and individual differences. Do not tease or shame learners about puberty-related changes.

• Any disclosure of abuse, coercion or a safety concern must be handled through the school's safeguarding and child-protection procedures.

• Questions beyond the lesson scope should be referred to an appropriate health professional or trusted adult.

Check your understanding

• Define reproduction and fertilisation.

• Name the male and female gametes.

• State the functions of the testes, ovaries, uterus and oviducts.

• Describe two changes that may occur during puberty.

• Explain the role of the placenta during pregnancy.

3.1 Transformation of Energy

Energy can be transferred from one object to another and transformed from one form to another. Learners identify energy changes in familiar devices and systems and consider efficiency and responsible energy use.

Forms of energy

• Common forms include kinetic, gravitational potential, elastic potential, chemical, electrical, thermal, light, sound and nuclear energy.

• Kinetic energy is associated with movement. Potential energy is stored because of position or configuration.

• Chemical energy is stored in fuels and food; electrical energy is associated with electric charges and circuits; thermal energy is related to the random motion of particles.

Energy transformation

• An energy transformation occurs when energy changes from one form to another.

• An electric lamp transforms electrical energy mainly into light and thermal energy.

• A battery-powered torch transforms chemical energy in the battery to electrical energy, then to light and thermal energy.

• A moving vehicle transforms chemical energy in fuel into kinetic energy, thermal energy and sound.

• A falling object transforms gravitational potential energy into kinetic energy as it falls.

Energy transfer and conservation

• Energy is conserved: it is not created or destroyed, but transferred or transformed from one form to another.

• In real devices, some energy is often transferred to the surroundings as thermal energy or sound that is not the intended useful output.

• Energy efficiency compares useful energy output with total energy input. It is commonly expressed as a percentage: efficiency = (useful energy output ÷ total energy input) × 100%.

• An efficient device wastes a smaller proportion of the input energy, although no real device is perfectly efficient.

Renewable and non-renewable sources

• Renewable sources are replenished naturally on a human timescale, for example solar, wind, hydroelectric and geothermal energy.

• Non-renewable sources are finite or take extremely long periods to form, for example coal, petroleum and natural gas.

• Energy choices have economic, environmental and social effects. Discuss local examples and responsible use.

Investigations and applications

• Trace energy transformations in a torch, a simple electric circuit, a solar-powered device or a swinging object.

• Draw energy-flow diagrams using arrows and labels.

• Compare energy-saving practices at home and school, such as switching off unused lights and using efficient appliances.

• Do not dismantle mains-powered appliances. Use low-voltage classroom circuits under supervision.

Check your understanding

• List four forms of energy.

• Describe the energy transformations in an electric fan and a torch.

• State the law of conservation of energy in your own words.

• Explain why a device may not be 100% efficient.

• Give two renewable and two non-renewable energy sources.

3.2 Pressure

Pressure describes how force is distributed over an area. Learners investigate pressure in solids and fluids, use the appropriate relationship and explain applications in everyday life.

Pressure in solids

• Pressure is force acting per unit area, when the force acts normally to the surface.

• Pressure = force ÷ area. The SI unit is the pascal (Pa), equivalent to one newton per square metre (N/m^2).

• For the same force, a smaller contact area produces greater pressure; a larger contact area produces lower pressure.

• Examples include sharp knives, drawing pins, wide vehicle tyres and snowshoes.

Calculations involving pressure

• Use $P = F/A$, where P is pressure in pascals, F is force in newtons and A is area in square metres.

• Rearrange the relationship when required: $F = P \times A$ and $A = F/P$.

• Convert area to square metres before substituting if the force is in newtons and the answer is required in pascals.

• Example: a force of 200 N acting normally on an area of 0.5 m^2 produces $P = 200/0.5 = 400$ Pa.

Pressure in liquids

• A liquid exerts pressure on the walls and base of its container.

• Liquid pressure increases with depth because of the weight of liquid above a point.

• At the same depth in the same stationary liquid, pressure is the same in all directions.

• Liquid pressure depends on the density of the liquid, gravitational field strength and depth: $p = \rho gh$ for the pressure due to a liquid column.

• Dams are made thicker near the bottom because water pressure is greater at greater depths.

Pressure in gases and atmosphere

• Gas particles move randomly and collide with surfaces. Their collisions exert pressure.

• Atmospheric pressure is caused by the weight of air above a surface and the collisions of air particles.

• Atmospheric pressure decreases with increasing altitude because there is less air above.

• Suction cups, drinking straws and syringes operate through pressure differences; they do not pull in the usual sense, but surrounding pressure helps push fluid.

Hydraulic systems

• Liquids are difficult to compress. Pressure applied to an enclosed liquid is transmitted through the liquid in all directions (Pascal's principle).

• Hydraulic brakes and lifts use pressure transmission to transfer force.

• In an ideal hydraulic system, pressure at the connected pistons is equal; a larger output piston area can produce a larger output force.

• Use diagrams or safe models to show how a small force can produce a larger force when acting through different piston areas.

Practical work and safety

• Investigate pressure in solids using blunt-ended objects and safe materials; do not use sharp blades on learners' skin.

• Demonstrate liquid pressure with a teacher-prepared plastic bottle with small holes at different heights, outdoors or over a sink.

• Use low-pressure, teacher-approved apparatus only. Never seal and heat a container or use compressed gas cylinders without trained supervision.

• Record measurements with units and show calculation steps clearly.

Check your understanding

• Define pressure and state its SI unit.

• Calculate the pressure when a force of 60 N acts on an area of 0.2 m^2.

• Explain why a sharp knife cuts more easily than a blunt one.

• Why are dams thicker at the bottom?

• Describe how a hydraulic lift uses pressure transmission.

End-of-year revision and integrated practice

Use these questions for recap, discussion, practical planning and formative assessment. They cover the eight Grade 8 Integrated Science substrands in this resource.

• Create a concept map linking elements, compounds, mixtures and physical or chemical changes.

• Explain the fire triangle and prepare a school fire-prevention and evacuation checklist.

• Draw and label typical plant and animal cells, then compare their structures.

• Use particle diagrams to explain diffusion and osmosis; distinguish these from active transport.

• Draw simplified male and female reproductive system diagrams and label the structures studied.

• Trace energy transformations in a solar lamp, an electric motor and a moving bicycle.

• Solve pressure problems using P = F/A and explain two applications of pressure in daily life.

• Plan a safe investigation: state the question, prediction, variables, apparatus, method, observations and conclusion.

Teacher reminder: Confirm the depth and exact scope of each lesson against the current KICD Grade 8 Integrated Science Curriculum Design and approved learner's materials. Use age-appropriate language, practical safety procedures and inclusive classroom discussion.